

Unit Rate Decision Tournament

Unit 4: Arcade Builder · Lesson 4-7 · EduWonderLab

UNIT 4 · LESSON 4-7 · TEACHER NOTES

Standard 6.RP.3b

FORMAT Bracket decision game

TIME 25–35 min

GROUPING Pairs or teams

TOPIC Solve Problems with Unit Rates

STANDARD 6.RP.3b

PREP LEVEL Light — print bracket and rate cards

Learning goal *I can use unit rates to solve real-world problems and choose the better option.*

Teacher setup

- Print one bracket and card set per pair.
- Students solve each rate problem and advance the better option.
- Continue until the best deal wins the bracket.

Materials

Tournament bracket, Rate cards, Explanation boxes, Pencil.

Differentiation & language support

- Explanation boxes prompt “I used unit rate because ____.” (SPED)
- Sentence stem: “I used the unit rate ____ to find ____.”
- ESOL: label every rate with its unit.

Key vocabulary (English / Español):

Term	Español	What it means
unit rate	tasa unitaria	a rate per one unit
constant rate	tasa constante	a rate that stays the same

Quick teacher check: *Ask a pair to explain the unit rate they used on Card 7.*

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Name: _____

Date: _____

Class: _____

Your goal: *I can use unit rates to solve real-world problems and choose the better option.*

What you need

Tournament bracket, Rate cards.

How to play

1. Read a rate card and find or use the unit rate.
2. Solve the problem.
3. Compare the two options and advance the winner.
4. Explain why you used the unit rate.

Rate Problem Cards

Use the unit rate to solve each problem, then advance the better option through the bracket.

1. A car drives 60 mph. How far in 4 hours?	2. A printer prints 20 pages per minute. Pages in 5 minutes?
3. Nuts cost \$0.25 per ounce. Cost of 12 ounces?	4. You run 3 miles in 24 minutes. At that pace, time for 5 miles?
5. A faucet fills 2 gallons per minute. Time to fill 30 gallons?	6. Job A pays \$8/hr, Job B pays \$9/hr. Earnings for 5 hours each?
7. A recipe uses 2 cups for 4 servings. Cups for 10 servings?	8. A heart beats 72 times per minute. Beats in 10 minutes?

Record your work (unit rate + answer):

Explain your thinking

Explain how a unit rate lets you solve for any amount, not just the one given.

Sentence frame: *I used the unit rate ___ per ___ to find ___.*

★ **Challenge:** *Card 4 found a running pace. How far could you run in 1 hour at that pace? (7.5 miles)*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. A car drives 60 mph. How far in 4 hours? Answer: 240 miles	2. A printer prints 20 pages per minute. Pages in 5 minutes? Answer: 100 pages
3. Nuts cost \$0.25 per ounce. Cost of 12 ounces? Answer: \$3.00	4. You run 3 miles in 24 minutes. At that pace, time for 5 miles? Answer: 40 minutes (8 min/mile)
5. A faucet fills 2 gallons per minute. Time to fill 30 gallons? Answer: 15 minutes	6. Job A pays \$8/hr, Job B pays \$9/hr. Earnings for 5 hours each? Answer: B = \$45 > A = \$40
7. A recipe uses 2 cups for 4 servings. Cups for 10 servings? Answer: 5 cups (0.5 cup/serving)	8. A heart beats 72 times per minute. Beats in 10 minutes? Answer: 720 beats

Teaching notes & look-fors

- Find the unit rate first, then multiply or divide for the new amount.
- Challenge: 8 min/mile → $60 \div 8 = 7.5$ miles in an hour.